

Season Performance Trends for Premier League Teams

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GitHub Link to Project: <https://github.com/conleyth-123/CSE482-Final-Project.git>

Abstract

I investigated the seasonal performance trends of teams in the English Premier League (EPL) by aiming to visualize the number of wins for each team over the ten seasons spanning from 2010 to 2020 and simultaneously building a match outcome predictor using this same dataset which included comprehensive match statistics such as the participating teams, the Full Time Result (FTR) indicating the final score and shot attempts (HS/AS). This analysis revealed the league is clearly divided into two subsections, top performers and bottom performers, a known divide that the visualization strongly showed. The top performers consistent success is often backed by significantly higher spending compared to bottom teams. The bottom teams frequently dropped in and out of the league due to the relegation system, while the top teams consistently reached a performance peak, slightly dropped and then returned to high performance levels. The match outcome predictor further confirmed that the top-performing teams consistently beat bottom-performing teams, although there are some cases where bottom teams win.

Intro

The project originated from a desire to combine my interest in machine learning with an interesting and relevant subject area that interest me. This led to the idea of creating a match predictor for the English Premier League (EPL) and thoroughly analyzing team performance trends across ten seasons spanning from 2010 to 2020. The analysis utilized a comprehensive dataset which included detailed match statistics from every match in the 10-year span. The research question was to determine if a universal, league-wide performance pattern exists across all teams, while also developing a model capable of predicting match outcomes.

Method

This project is built around analyzing historical season and match data, using a chart and a skill rating system that feeds a machine learning model for match prediction. This project used Flask, Pandas, NumPy, scikit-learn and joblib. Below is a breakdown of the four major steps.

1. Getting the Data Ready and Calculating Skill Ratings (Elo)

We first had to load all the raw data, which was 10 CSV files of each season, and then stitch them together into one massive timeline.

Data Preparation: I start by loading all the match records from multiple seasons and combining them into a single, comprehensive dataset. The most important thing is

chronological sorting. I want to process the matches in the exact order they were played, because the entire system relies on constantly updated skill ratings.

The Elo System: The core feature of the entire project is the Elo rating. This is a skill score for each team that adjusts after every match. Every team starts at the same base skill level of 1500, and it moves from there.

- Home Field Advantage: Before any home game, the home team gets a small temporary boost of 100 points to their rating, this accounts for the natural advantage of playing at their own stadium with the fans.
- Prediction vs. Reality: I use the current Elo ratings to calculate the probability of a team winning.
- Updating: Once the match is finished, a key formula is used to update the teams' ratings. If a team wins against a stronger opponent, which mean a higher Elo, they gain a lot of points. If they lose to a weaker opponent, they will lose a lot of points. This makes the Elo rating an incredibly sensitive measure of a team's current form and skills. This updated Elo rating is then stored and used as the main predictive input for the next match.

The formula used to calculate the new Elo rating for a team ($R_{A_{new}}$) is:

$$R_{A_{new}} = R_A + K * (S_A - E_A)$$

$$\text{where } E_A = \frac{1}{1 + 10^{\frac{R_B - R_A}{400}}}$$

2. Making Overall Stats

Once the data is all loaded and processed, I run a quick analysis to get some context for the entire league across all seasons. This gives us the big picture, and a high-level summary that sets the scene:

- Total Data: The total number of matches and seasons I looked at.
- Scoring Averages: The average number of goals scored per match across the whole dataset.
- Win Rates: The overall win percentage for home teams versus away teams.

3. Season Performance Trend by Team: Total Wins Across Seasons

To truly see how consistent a team has been over the years, I created a chart to track their total number of wins per season. I don't just look at their skill now, but their historical long-term performance and find any patterns.

The Counting Process:

1. Tagging Wins: I filter the data to identify every match that resulted in a Home Win and every match that resulted in an Away Win.
2. Grouping: For every single team, I add up their total home and away victories in the same season.
3. Combining and Finalizing: I then combine these two counts for every season to calculate the total wins. If a Team has zero wins at home or away it still marks it as 0 to ensure a data point isn't missing, this is crucial for if a team gets relegated and is no longer in the Premier League.

The Output: This process results in a clean, structured record of every team's total wins for every season. This structure makes it easy for the visualization part of the project to plot a historical trend line for any team you select.

4. Building the Machine Learning Model

The final step is building the Machine Learning Model that makes the predictions. I use the Elo ratings I calculated in Step 1 as the inputs/features for the model.

Features for Prediction:

- Elo Difference: The most important input is the difference between the two teams Elo ratings. The home team gets an extra boost.
- Probability: The model also uses the Elo to calculate the probability of a home team win, away team win, or a draw.
- Raw Scores: The raw Elo scores themselves are also included, which helps the model, and I display it so the user can understand.

Training the Model: I used a Logistic Regression model, which is a common and an effective tool for classifying things, in my case, classifying the match result as a Home Win, a Draw, or an Away Win. The raw feature numbers are first standardized so the model can learn efficiently, and then model is trained on the historical data.

Prediction and Fallback:

- When you ask for a prediction, the model takes the two teams current Elo scores, does the math and gives you three main outputs, the most likely final score, the confidence level, and the probability for a Home Win, Draw, and Away Win.
- I also included a system to predict a most likely final score based on the Elo difference.
- If the main, trained model file is ever missing or can't be loaded, the system automatically uses Elo based prediction logic as a backup. This ensures the model always works.

Results

Across the ten seasons analyzed (2010/11 to 2019/20), the dataset included a total of 3,628 matches, providing a strong foundation for identifying long-term patterns. During this period, teams averaged 1.38 points per game and scored an average of 2.75 goals per match. Home teams held a clear advantage with a 45.6% win rate, compared to a 29.7% win rate for away teams. These overall statistics help contextualize the performance trends observed throughout the analysis.

As I was analyzing the data and wins, I began to notice two major subgroups forming: top-performers and bottom-performers. Almost every team fell into one of these categories, though a few were more middle-of-the-pack. I will first discuss the bottom-performers. A pattern I noticed among these teams was that they were constantly fighting for a place in the league. Because of the league's relegation system, teams can drop out and then return in later seasons. These bottom-performers frequently moved in and out of the league. They often performed just well enough to earn promotion but would either drop out the following year or gradually decline as their best players left for stronger, top-performing clubs. Since they are constantly battling to stay in the league, their performance lines tended to look like small hills, rising from zero (when relegated), gaining a few wins, and then falling back to zero again.

In comparison, we have the top-performing teams. These are the clubs that attract talented players from the bottom-performing teams, which in turn widens the performance gap. Top-performing teams, such as Liverpool, Manchester United, and Chelsea never had to fight to remain in the league. While they occasionally had a poor season, they were never at risk of relegation. A common trend I found in their win charts was that after a strong season with many victories, they often performed worse the following year. I found this interesting because it appeared consistently across all top performers. There could be many reasons for this but based on my experience with the team I personally support, it's common for star players to leave or be transferred to other major leagues, such as La Liga.

Also, when comparing top performers and bottom performers in the match outcome predictor, the results were very consistent. Nearly every time the predictor favored the top-performing team with a high level of confidence. In cases where it did not project a clear win, the model typically predicted a draw rather than a loss for the stronger team. This pattern reinforces the performance divide observed in the data: top performers not only accumulate more wins over a season, but they also hold a measurable statistical advantage in matchups. Even when the model accounted for variables such as home-field advantage, recent performance, and goal differences, the gap between the two groups remained significant. This suggests that the differences between top and bottom teams, such as squad depth, financial resources, and player quality, on average directly translate into predictable match outcomes.

Lesson Learned

When working on this project, I learned about applying machine learning to a real-world dataset, which was something I was excited to explore. I gained hands-on experience cleaning, preparing, and analyzing a large set of match data, and it helped me understand how statistical patterns translate into meaningful models. One of the most interesting things I learned from my results was how clearly the data revealed a divide between top-performing and bottom-performing teams. Seeing how consistently top performers outmatched lower-tier teams, both in long-term trends and in the match predictor, showed me the power of data to highlight differences within a league. Beyond the analysis, I also expanded my knowledge on how to integrate these findings into a functional web app, giving me practical experience in both machine learning and software development. Overall, this project strengthened my technical skills and helped me appreciate how data-driven approaches can uncover patterns that aren't always obvious from just watching the games.

Future Improvements

Some future improvements I would make include adding a section that allows users to dive deeper into a single season. This feature would display detailed performance metrics for each team during that specific year, giving a more focused view of how a team developed and performed over time. Building on that, it would also be interesting to integrate this into the match predictor so users could compare teams across different seasons. For example, someone could match a historically bottom-performing team like Leicester City during their miracle 2015/16 title run against a top-performing team like Chelsea in one of their championship seasons to see who the model predicts would come out on top. This would add a fun, interactive “what-if” element and make the analysis more engaging for users.